# Module 2: Infrastructure Security

# Topic 3: Vulnerability Assessment and Penetration Testing

# Vulnerability Assessment

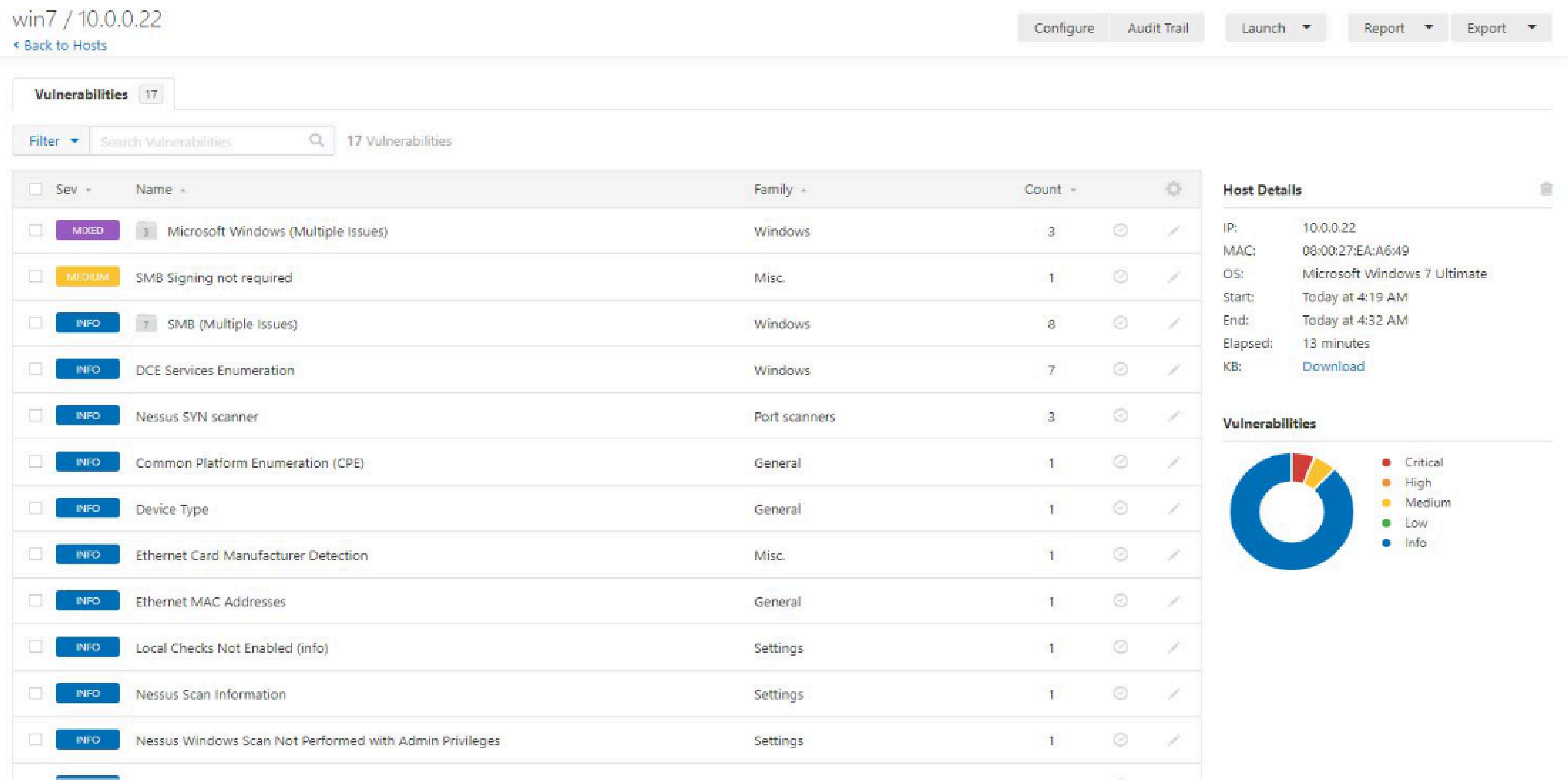
A vulnerability assessment is the process through which we identify the weak points that could be exploited on your network, along with how important the associate threat really is. Silent Breach conducts a thorough analysis to determine the attack surface that is intentionally or unintentionally exposed, and correlates it with a risk value to determine your security posture.

# Vulnerability Assessment

- Tools:
  - OpenVAS
  - **Nessus**
  - Nexpose
  - Outpost24
  - Qualys

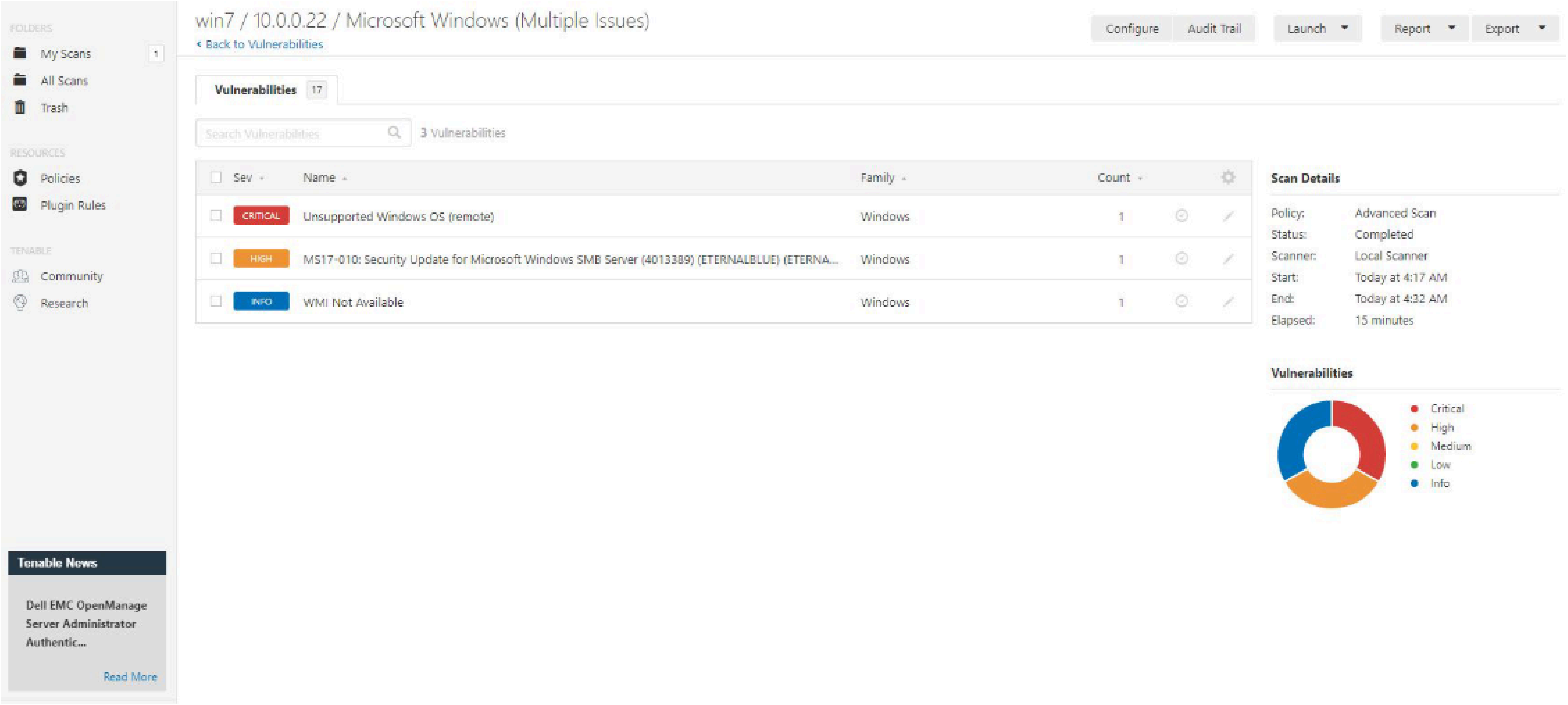
# Vulnerability Assessment

## Nessus



# Vulnerability Assessment

## Nessus



# Vulnerability Assessment

## Reporting

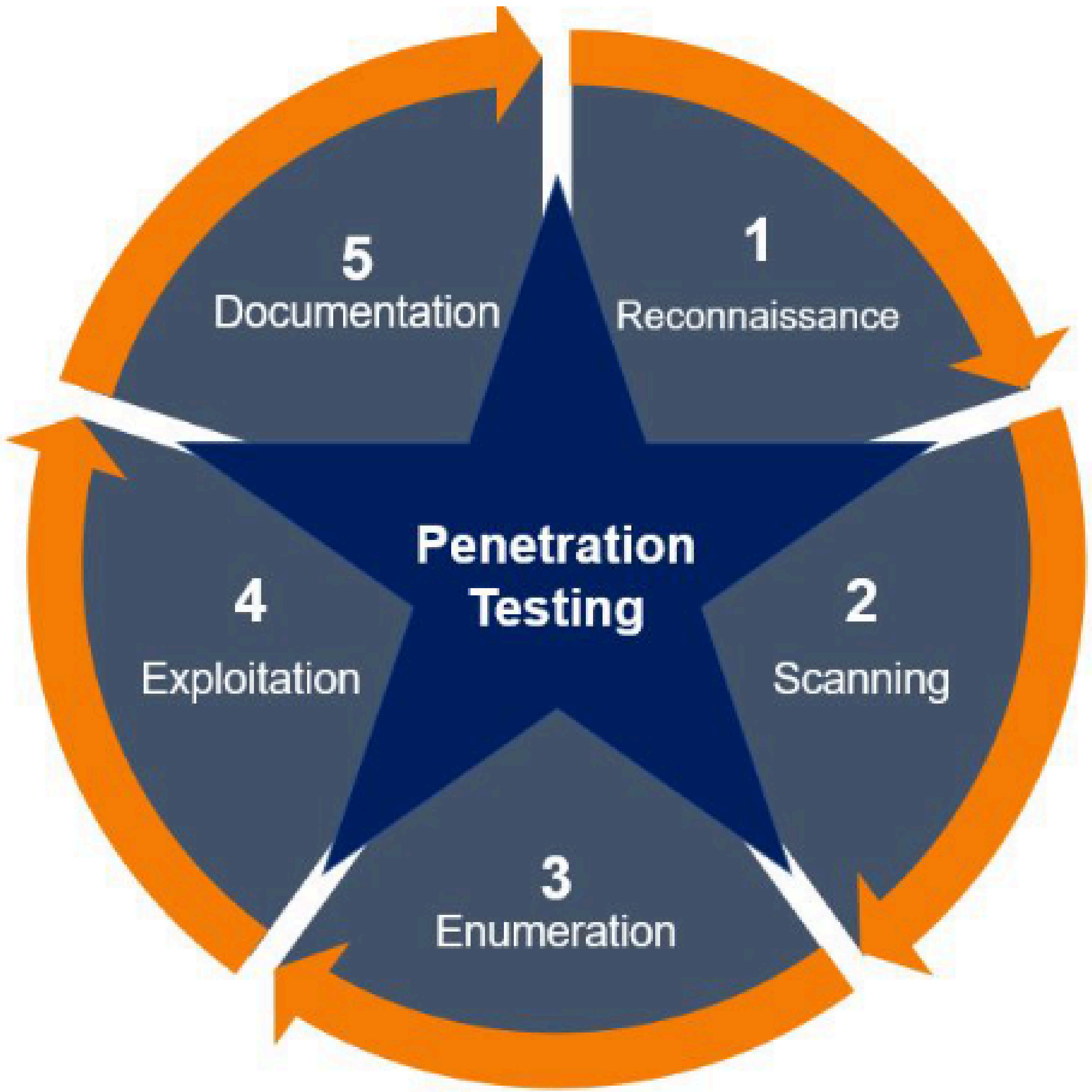
| High Level Report | Vulnerability Report | Step by Step Guide |
| --- | --- | --- |
| Statistics | Each IP with Vulnerability | Starting with NMAP how to Scan |
| Overview of Network | Contain Exploit Details | Trail of Thoughts what we were thinking |
| How Secure the network | Screenshot of Successful Exploit | Fail Exploits |
| Works Well for Management | Screenshot of Privilege Escalation | Trial and Errors |
| | How To Fix the Vulnerability | |

# Introduction to Penetration Testing

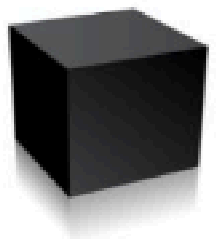
**Penetration testing** (also called **pen testing**) is the practice of **testing** a computer system, network or Web application to find vulnerabilities that an attacker could exploit.

# Introduction to Penetration Testing

- Information Gathering / Reconnaissance
- Scanning
- Enumeration
- Exploitation
- Documentation and reporting

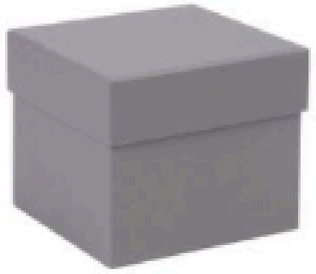


# Types of Penetration Testing



**Black Box**

- Zero knowledge
- No access /credentials given
- Test as external attacker



**Gray Box**

- Partial knowledge
- Minimum access or credentials given
- Test as normal user



**White Box**

- Full knowledge
- All access and credentials given
- Test as developer

# Why Penetration Testing?

1- Find vulnerable systems before hackers do.

2- Determine strength of security controls used in environment.

3- Prioritize security investments and security controls needed.

4- Adhere to some regulatory or framework requirements.

5- Protect corporate image and brand reputation.

# Vulnerability Assessment Vs Penetration Testing

| Vulnerability Assessment | Penetration testing |
|---|---|
| Automated activity | Manual Activity |
| Low price (price dependent on used tool) | Expensive (depending on experience of pentester) |
| Takes few time to be completed | Takes long time |
| Can be done on large scale | Limited scope |
| Detect vulnerabilities without exploitation | Exploit available vulnerabilities |
| Multiple false positives | Rare false positives |